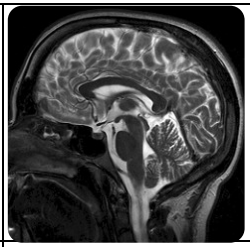
Ways to Study the Brain

Complete the following chart based on ways we study the brain. The "way" is listed…add a description of how it is used and a visual.

Term	Explanation	Visual
Lesioning/Injury/Damage	Lesions allow the scientist to observe any loss of brain function that may occur. Psychologist may cause a lesion on animals or people to study how the damage or lesion can affect the people behavior.	
Electrical Stimulation	To place detectors in the brain to study the responses of specific neuron. Primarily used in animals	
EEG	A less invasive approach, and one that can be used on living humans. A technique that records the electrical activity produced by the brain's neurons through the use of electrodes that are placed around the research participant's head.	

CT Scan	Computer tomography : builds up a picture of the brain based on the differential absorption of X-rays. Reveal the gross features of the brain but do not resolve its structure well.	
MRI	provides more specific brain images. type of brain scan that uses a magnetic field to create images of brain activity in each brain area. The fMRI detects the amount of blood flow in each brain region, and thus is an indicator of neural activity.	
PET	Positron Emission Tomography (PET) uses trace amounts of short-lived radioactive material to map functional processes in the brain.	

The 3 imaging techniques are:

CT scan
MRI scan
PET scan